

DEG-KDA Results and Findings



Eric Zhuang
8/4/2026



Experiment Setup & Methodology

Multi-layered transcriptomic profiling across fine cell types, sex, and APOE strata



General Setup

- ❑ DEG analysis: per fine cell type and per sex/APOE
- ❑ Pathway: 4264 pathways
- ❑ KDA parameters and setup:
 - ❑ 3 layers to try and pick the best
 - ❑ For each signature, passed the network for the corresponding broad cell type
 - ❑ Background is the nodes in the cell-type network

Signatures in Primary and Secondary Runs for KDA

The methodology executes a comprehensive **1,782 planned runs**, strategically dividing the workload between 972 primary strata and 810 secondary pooled summaries to ensure robust statistical coverage.

Primary and Secondary Runs

1,782 planned runs

PRIMARY

six individual strata

F e2

F e33

F e4

M e2

M e33

M e4

each AD-NCI contrast remains separate

▲ AD-up

▼ AD-down

◆ Both

$D_{\text{both}} = D_{\text{up}} \cup D_{\text{down}}$

$54 \times 6 \times 3 = 972$

SECONDARY

five pooled summaries

Female
F e2+e33+e4

Male
M e2+e33+e4

e2
F+M

e33
F+M

e4
F+M

▲ P-up

▼ P-down

◆ Both

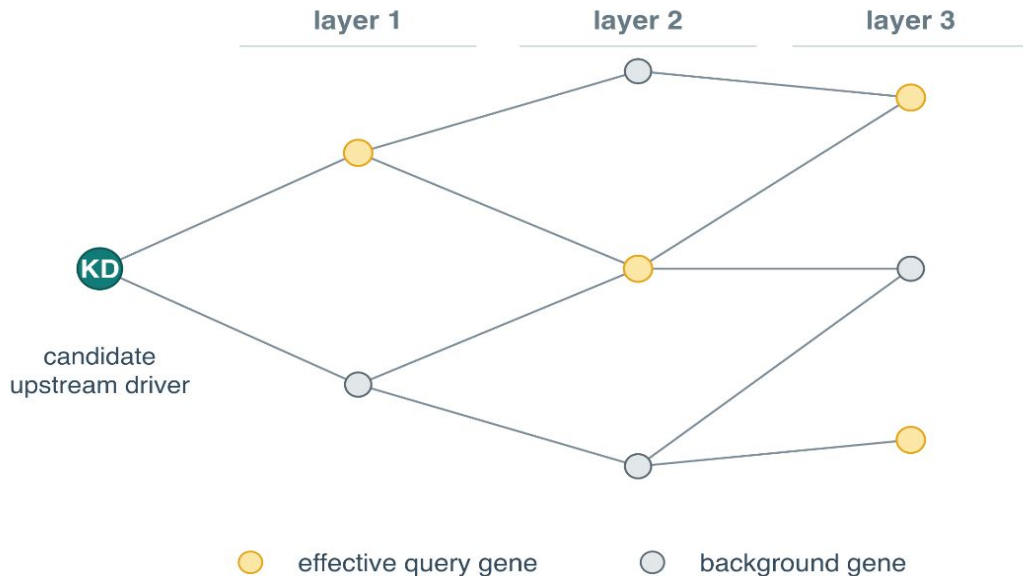
DEG set unions • no pooled model refit
direction discordance retained + flagged

$54 \times 5 \times 3 = 810$

KDA Setup

The setup identifies putative key drivers by evaluating fold enrichment using a **one-sided hypergeometric test**. Drivers are successfully reported when the **adjusted P-value is ≤ 0.05** .

NetWeaver Key Driver Analysis



Is the query concentrated downstream?

M background genes

m neighborhood genes

k query genes

q query genes in neighborhood

$$\text{fold enrichment} = (q/m) / (k/M)$$

one-sided hypergeometric test
BH correction across drivers within this run

adjusted P ≤ 0.05

directed • layers 1–3 • no signature expansion • overlap genes returned

REPORT

Putative key driver
best layer • overlap • enrichment • adjusted P

No driver passes cutoff
= completed, not failed



Mitochondrial DEG Analysis Results

Uncovering sex-specific expression dynamics and OXPHOS regulatory changes

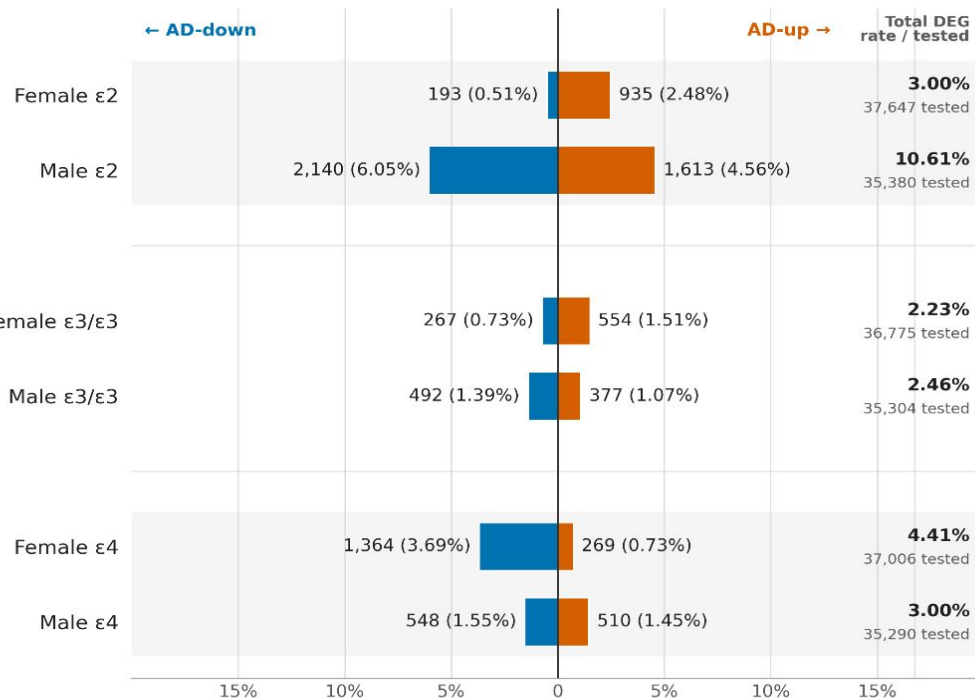


Direction and burden of AD-associated mitochondrial transcription

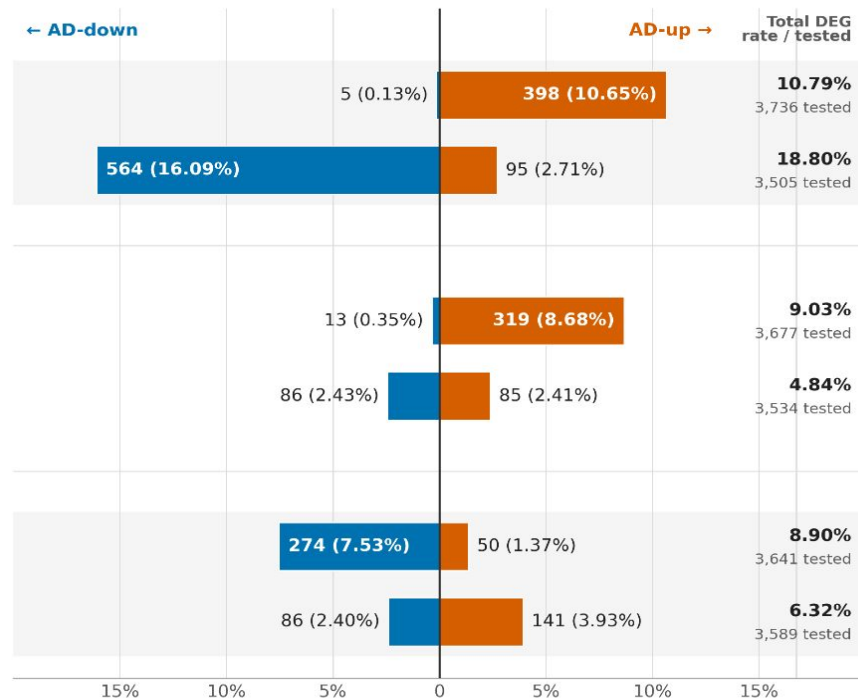
AD versus no cognitive impairment, stratified by sex and APOE genotype

AD-down AD-up

All mitochondrial genes



OXPHOS subunits



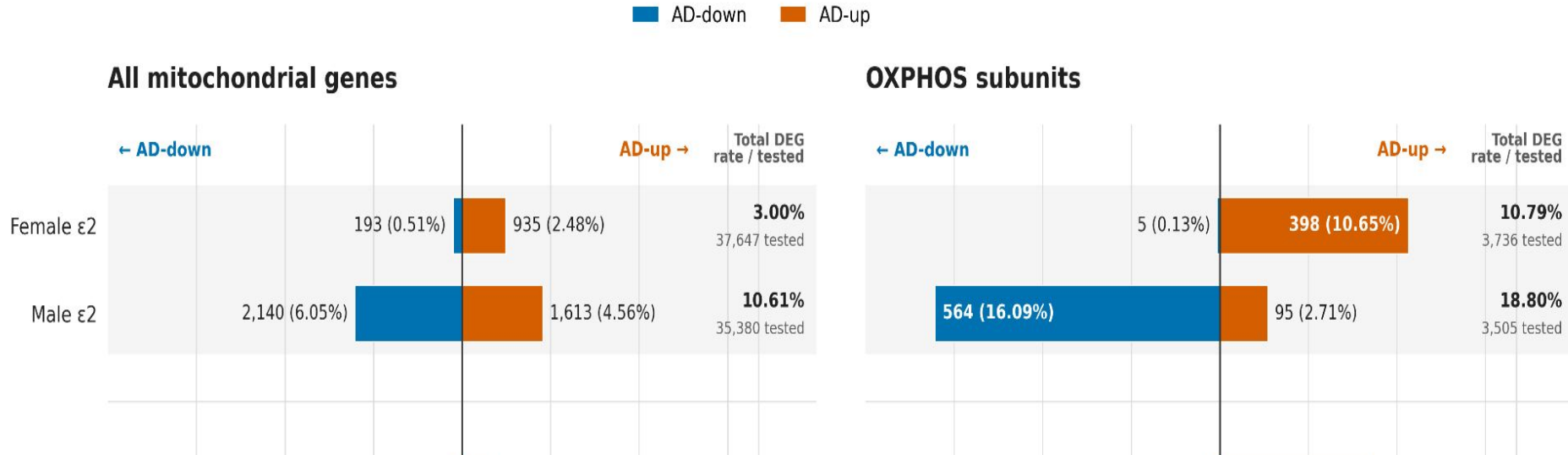
Significant gene-fine-cell-type occurrences (% of all tested occurrences)

Bar labels are occurrence count (percentage tested). Total DEG rate is the combined AD-up and AD-down percentage.

Occurrences can repeat a gene across fine cell types and are not unique genes or donors; strata are descriptive, not formal interaction tests.

Direction and burden of AD-associated mitochondrial transcription

AD versus no cognitive impairment, stratified by sex and APOE genotype



Key Finding:

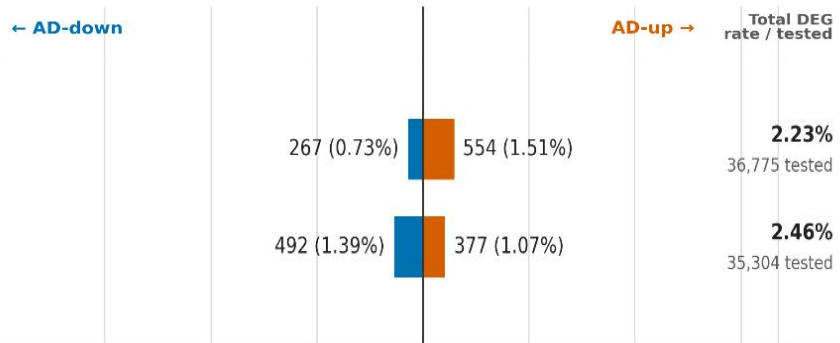
General upregulation in female ε2, but downregulation in males (sex reversal)

Direction and burden of AD-associated mitochondrial transcription

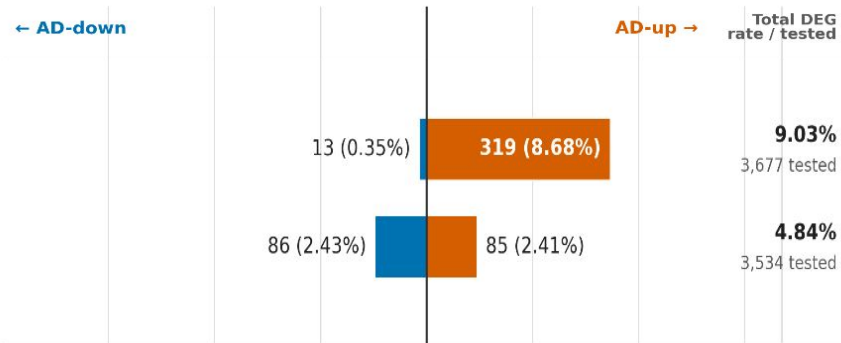
AD versus no cognitive impairment, stratified by sex and APOE genotype

AD-down AD-up

All mitochondrial genes



OXPHOS subunits

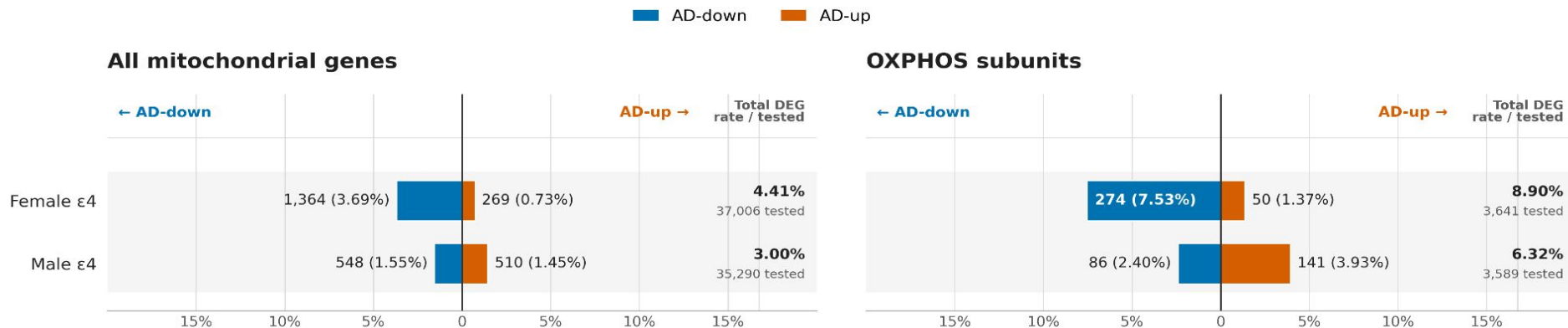


Key Finding:

Mixed/inconclusive results for e3/e3 and male e4z

Direction and burden of AD-associated mitochondrial transcription

AD versus no cognitive impairment, stratified by sex and APOE genotype



Key Finding:

Downregulation of female e4

General expression trends (comparison with literature)

Support for APOE4 regulatory change:

- ❑ Reduce respiration and ATP capacity (Farmer et al., 2021)
- ❑ Impair fatty-acid support between astrocytes and neurons (Qi et al., 2021)
- ❑ Disrupt mitochondria/mitophagy (Lee et al., 2023).
- ❑ The female AD cortex has also shown lower Complex IV respiration (Yang et al., 2025).
- ❑ Sex reversal in APOE e2 is mostly unsupported by prior study.

Differential expression of OXPHOS Pathway genes



Stratum	OXPHOS occurrences / tested	AD up	AD down
Female $\epsilon 2$	403 / 3,736 (10.79%)	398	5
Female $\epsilon 3/\epsilon 3$	332 / 3,677 (9.03%)	319	13
Female $\epsilon 4$	324 / 3,641 (8.90%)	50	274
Male $\epsilon 2$	659 / 3,505 (18.80%)	95	564
Male $\epsilon 3/\epsilon 3$	171 / 3,534 (4.84%)	85	86
Male $\epsilon 4$	227 / 3,589 (6.32%)	141	86

OXPHOS Regulation

Strong, robust changes in the regulation of oxidative phosphorylation genes are predominantly localized within excitatory neuronal populations

Electron Transport Chain

Pathway alterations are driven heavily by electron transport chain proteins, with particular enrichment among Complex IV subunits.



Key Driver Analysis (KDA) Findings

Three major regulatory hubs driving mitochondrial and metabolic pathology in AD



1. Astrocyte APOE regulates TUFM and ATP synthase

Astrocyte APOE regulates TUFM and ATP synthase

- ❑ APOE has been returned the key driver in 20 (7 primary)
- ❑ All primary calls are in runs on downregulated set
- ❑ Each time, it regulates TUFM directly or indirectly
- ❑ ATP5PB is regulated in 13, while ATP5F1A in 6
- ❑ Female e2, Male e2 and Male e4
- ❑ Median fold enrichment of 17.3

Astrocyte APOE regulates TUFM and ATP synthase

- ❑ This appears in all astrocyte fine cell types, strongest in GRM3

Astrocyte fine cell type	APOE/TUFM P/S/total	Any ATP5 subunit P/S/total
Ast CHI3L1	2/3/5	0/2/2
Ast DPP10	1/4/5	1/4/5
Ast GRM3	4/6/10	2/4/6

- ❑ TUFM appears 19 times, across many cell types and primarily in downregulated cases

Astrocyte APOE regulates TUFM and ATP synthase

Prior research:

- ❑ Connection between TUFM and AD (Zhong et al., 2021)
- ❑ APOE e4 reduces/impairs respiratory capacity (Qi et al., 2021) and oxidative phosphorylation (Lee et al., 2023)

2. LAMTOR5 connects nutrient sensing to ATP5IF1



LAMTOR5 connects nutrient sensing to ATP5IF1

- ❑ LAMTOR5 is a key subunit of the ragulator complex, involved in nutrient sensing
- ❑ LAMTOR5 is an upstream driver in 96 runs, 38 primary
 - ❑ 62(14.8% of excitatory) and 34(9.8% of inhibitory)
 - ❑ Span 15 cell types
 - ❑ All calls occur in neuronal subtypes

LAMTOR5 connects nutrient sensing to ATP5IF1

- ❑ ATP5IF1 appear regulated by LAMTOR5 70 times
- ❑ In all sex-APOE strata; appears only in upregulated genes in female $\epsilon 2$ and only in downregulated elsewhere

LAMTOR5 connects nutrient sensing to ATP5IF1

- ❑ ATP5MC2 has 202 calls, 56 in LAMTOR5 neighborhoods
- ❑ Few calls of any gene are global
- ❑ ATP5IF1 encodes inhibitory factor subunit 1 (IF1), which prevents wasteful hydrolysis of ATP

Tier and group	ATP5IF1	ATP5MC2
Primary female ϵ 2	8	8
Primary female ϵ 4	6	5
Primary male ϵ 2	14	10
Secondary female pool	10	9
Secondary male pool	8	8
Secondary ϵ 2 pool	19	13
Secondary ϵ 4 pool	5	3

LAMTOR5 connects nutrient sensing to ATP5IF1

- ❑ Regulator–mTORC1 nutrient sensing and neuronal ATP5IF1 biology are supported separately (Bar-Peled et al., 2012; Esparza-Moltó et al., 2021)
- ❑ No prior research supports a connection between the two complexes in Alzheimer's disease

3. GABARAPL2 connects autophagic regulation to CHCHD2 and PARK7

GABARAPL2 connects autophagic regulation to CHCHD2 and PARK7

- ❑ GABARAPL2; responsible for autophagy
- ❑ 79 calls, 32 of which are primary
 - ❑ Almost all (77) are in excitatory neurons; other 2 in inhibitory
 - ❑ Distributed among all sex-APOE groups

Tier	Group	GABARAPL2 Calls
Primary	Female ε2	7
Primary	Female ε3/ε3	0
Primary	Female ε4	5
Primary	Male ε2	13
Primary	Male ε3/ε3	6
Primary	Male ε4	1

GABARAPL2 connects autophagic regulation to CHCHD2 and PARK7

- ❑ **CHCHD2 occurs in 66 calls (25 primary, 41 secondary) across 11 fine cell types**
- ❑ **PARK7 occurs in 49 calls (16 primary, 33 secondary) across 7 fine cell types**
- ❑ **Occur together in 42 calls, all excitatory (13 primary, 29 secondary)**
- ❑ **Of the 79 calls, 24 contain CHCHD2 alone, 7 contain PARK7 alone, 42 contain both, and 6 contain neither.**

GABARAPL2 connects autophagic regulation to CHCHD2 and PARK7

Prior Research:

- ❑ GABARAP-family proteins are established regulators of autophagosome-lysosome fusion in autophagy (Wang et al., 2015)
- ❑ CHCHD2 is essential in regulating OXPHOS, apoptosis and stress response
- ❑ PARK7 protects against oxidative stress
- ❑ A GABARAPL2-dependent pathway can also confer benefit in AD-relevant models, improving amyloid, synaptic and behavioral phenotypes in mice (Dai et al., 2025).

Circular Figure

Network color
(outer sectors and R bars)

- Astrocytes
- Vasculature
- Excitatory neurons
- Inhibitory neurons
- Microglia
- Oligodendrocytes
- OPCs

T composition color

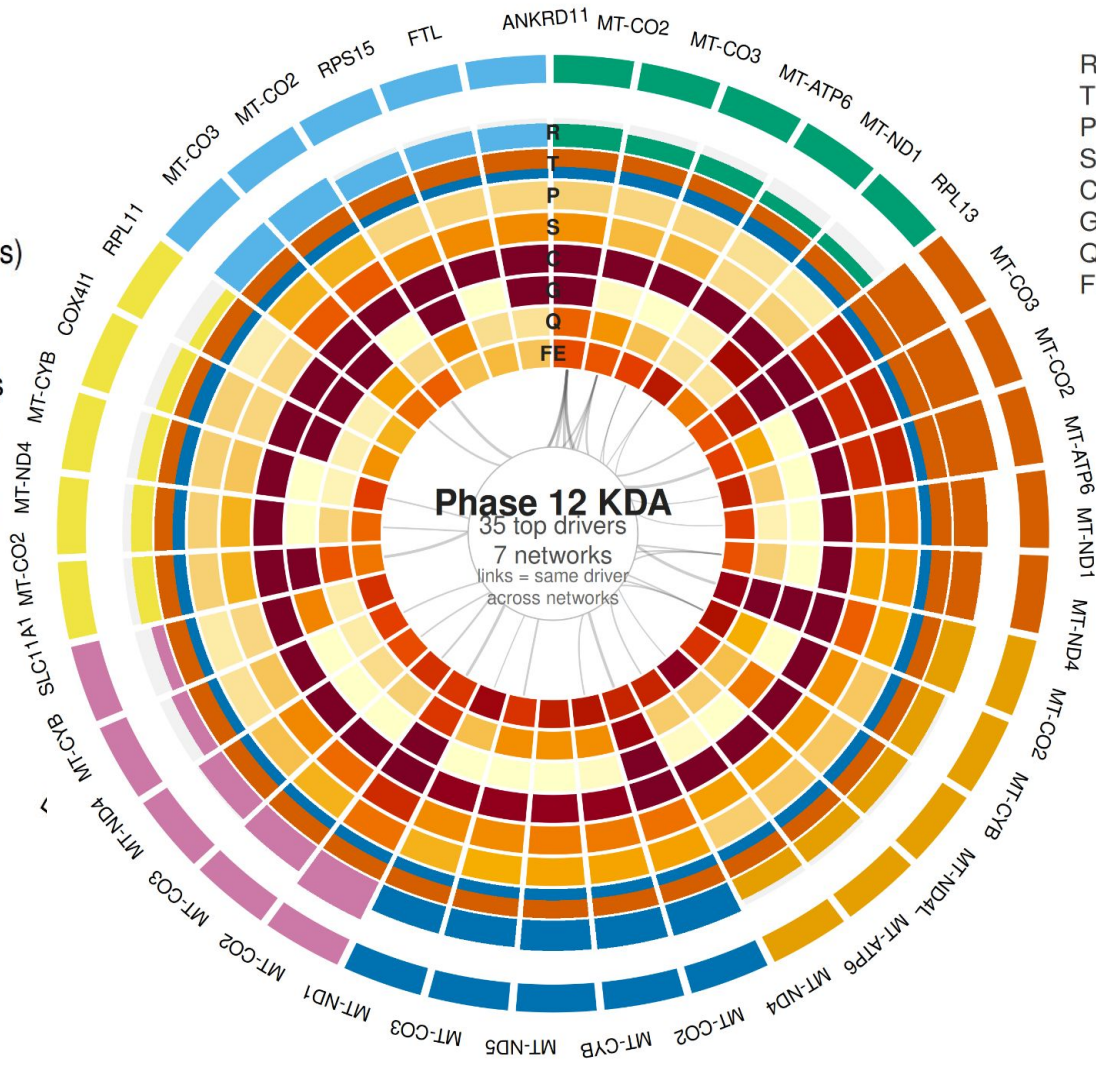
- Primary calls
- Pooled calls

Heat-ring color
(P, S, C, G, Q, FE)

- Low (0)
 Middle (0.5)
 High (1)

Tracks, outer to inner

- R all-run recurrence
T primary / pooled share
P primary recurrence
S pooled recurrence
C eligible cell-type coverage
G global-driver call share
Q min BH FDR strength (cap 25)
FE max fold-enrichment strength (cap 12)



4. Works Cited



- Bar-Peled L, Schweitzer LD, Zoncu R, Sabatini DM. Ragulator is a GEF for the Rag GTPases that signal amino acid levels to mTORC1. *Cell*. 2012;150(6):1196–1208. doi:10.1016/j.cell.2012.07.032.
- Dai X, Ye Z, Wang C, et al. SIK2-mediated phosphorylation of GABARAPL2 facilitates autophagosome–lysosome fusion and rescues neurodegeneration in an Alzheimer's disease model. *Translational Neurodegeneration*. 2025;14:53. doi:10.1186/s40035-025-00514-4.
- Esparza-Moltó PB, Romero-Carramiñana I, Núñez de Arenas C, et al. Generation of mitochondrial reactive oxygen species is controlled by ATPase inhibitory factor 1 and regulates cognition. *PLOS Biology*. 2021;19(5):e3001252. doi:10.1371/journal.pbio.3001252.
- Schmukler E, Solomon S, Simonovitch S, et al. Altered mitochondrial dynamics and function in APOE4-expressing astrocytes. *Cell Death & Disease*. 2020;11:578. doi:10.1038/s41419-020-02776-4.
- Wang H, Sun HQ, Zhu X, et al. GABARAPs regulate PI4P-dependent autophagosome:lysosome fusion. *Proceedings of the National Academy of Sciences of the United States of America*. 2015;112(22):7015–7020. doi:10.1073/pnas.1507263112.
- Williams HC, Farmer BC, Piron MA, et al. APOE alters glucose flux through central carbon pathways in astrocytes. *Neurobiology of Disease*. 2020;136:104742. doi:10.1016/j.nbd.2020.104742.
- Zhong B-R, Zhou G-F, Song L, et al. TUFM is involved in Alzheimer's disease-like pathologies that are associated with ROS. *The FASEB Journal*. 2021;35(5):e21445. doi:10.1096/fj.202002461R.
- Zhou W, Zhang MM, Tang W, et al. CHCHD2 and CHCHD10 promoted autophagic clearance of protein aggregates via GABARAPs. *Autophagy*. 2026:1–30. doi:10.1080/15548627.2026.2678427.